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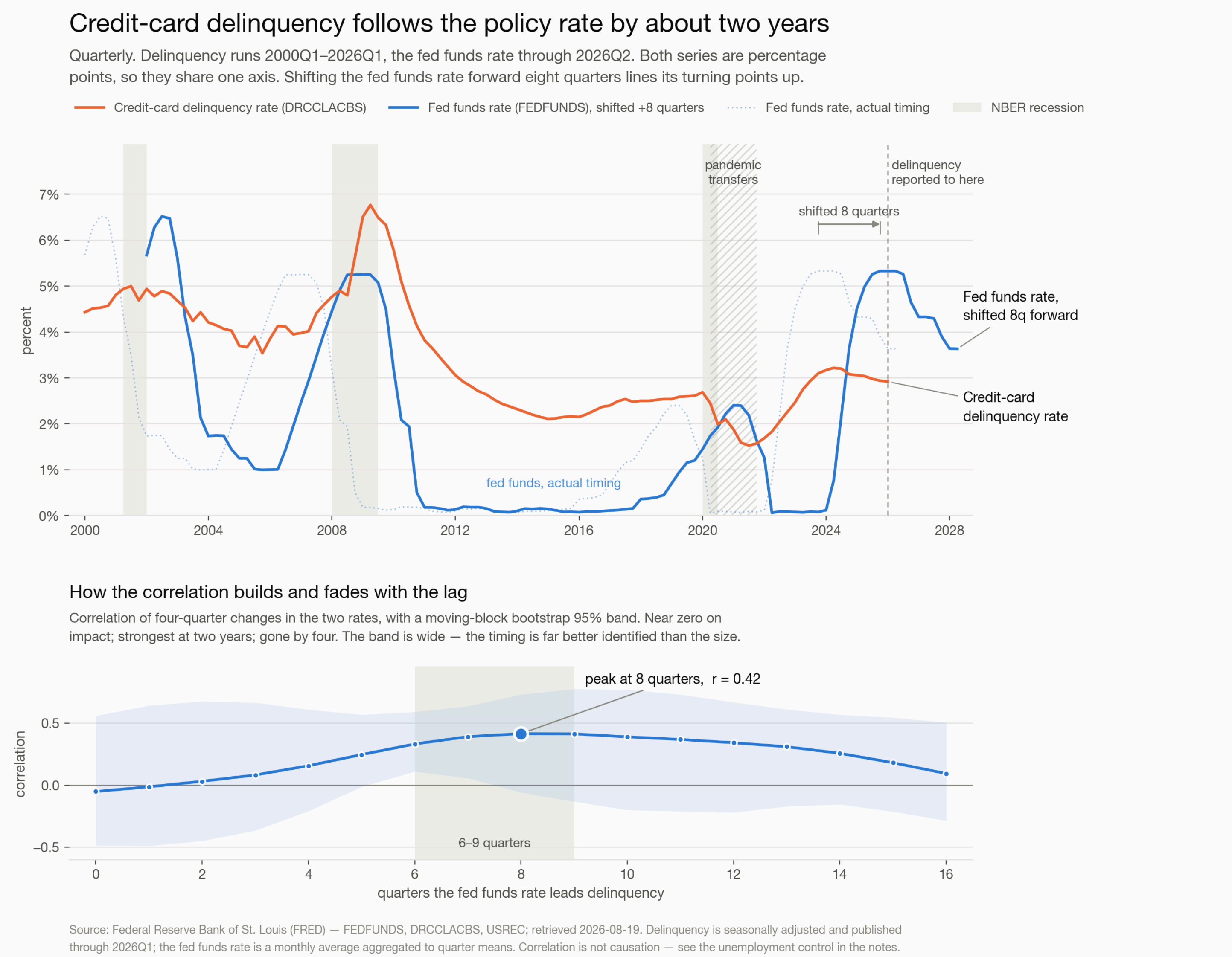
How Fed policy rates flow through to credit-card delinquency

Executive readout · credit-risk data science Generated in a Claude Code session on 2026-08-19 · data retrieved from FRED that day



Fed funds rate versus credit-card delinquency, with the lag annotated

The finding

Credit-card delinquency lags policy-rate changes by roughly two years — eight to ten quarters. In the same quarter, the two series are effectively uncorrelated. A risk team reading current-quarter rates against current-quarter delinquency is reading noise.

What was measured

Series	`FEDFUNDS` — effective fed funds rate, monthly, through July 2026
	`DRCCLACBS` — credit-card delinquency rate, all commercial banks, quarterly, through 2026 Q1
Window	2000 onward, aligned to 105 quarters
Method	Cross-correlation of year-over-year changes with bootstrapped 95% confidence bands, a distributed-lag regression, a Granger causality test, and a per-episode table of rate-cycle peak → delinquency peak

Results

- **Peak correlation at 8 quarters: r = 0.42** (95% CI 0.23 – 0.57).
- **Pre-2020 the relationship is markedly stronger: r = 0.71 at 9 quarters.** The pandemic period is the reason the full-sample figure is lower — transfer payments suppressed delinquency while rates moved, and that window is marked on the chart rather than dropped.
- **The 2007–09 episode is textbook:** the rate cycle peaked 2007 Q1, delinquency peaked 2009 Q2 — nine quarters.
- **Granger causality was not significant.** Stated here because it was run, not buried because it disagreed. The lag structure is a robust *association*; this analysis does not establish that rate changes cause delinquency changes.

What follows for a risk team

The practical consequence is about *timing*, not direction. A tightening cycle's effect on card portfolios shows up roughly two years later, which means:

- Provisioning models keyed to same-quarter rates are keyed to the wrong quarter.
- The 2022–24 hiking cycle's delinquency consequence had already arrived by the 2024 peak visible on the chart.
- Any forward view built on this relationship inherits its confidence interval — r = 0.42 with a band from 0.23 to 0.57 is a directional signal, not a forecast.

Provenance and honesty about this document

The chart is the actual output of the 2026-08-19 run. The figures above are transcribed from that run's recorded results — they were not recomputed when this page was assembled, and the analysis code is not preserved here because the workspace it ran in is wiped between demonstrations by design.

Correlation is not causation, the sample is one country over one 26-year window, and the pandemic period is a genuine structural break rather than an outlier to be discarded. Treat this as a well-evidenced starting point for a risk conversation, not as a model.
